Machine Learning Project Report

Using Machine Learning to Improve the Business of a Chain of Building Painting Works

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Group Number: 3

Course: Machine Learning (CSC-505)

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Name	**Enrolment Number**	**Contribution**
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Introduction

A large part of construction and maintenance of buildings is the painting of these buildings. Accurate cost estimations are vital for painting contracting businesses so they can produce accurate quotation, budget and work efficiently. If an inaccurate estimation is made they will lose money or have a quotation that doesn't compete.

Machine Learning (ML) can be applied to help these businesses understand the real cost by taking into consideration the relationship between previous spending and current project data. This project uses an open dataset relating to construction and electrical Bills of Quantities (BOQ) from Kaggle [1]. The data related to painting is extracted and different attributes of items and projects used to predict the cost of painting. Linear Regression, Decision Tree and Support Vector Regression (SVR) models are used to predict the cost. These models are compared based on their Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and coefficient of determination (R^2). The SVR performs best on the test set for all three metrics so is chosen as the winning model.

1.1 Problem Statement

The objective of this project is to develop a machine learning model to predict painting costs based on data from the items and construction projects they belong to.

During the course of this study, we will attempt to answer the following questions:

Is it possible to predict painting costs using historical BOQ data? Which of the three regression models performs best? Does a non-linear kernel model, such as SVM provide any significant benefit over a simple linear model?

1.2 Project Objectives

The main objectives are:

1. Obtain and preprocess a suitable public dataset.
2. Identify painting-related records and relevant input features.
3. Train Linear Regression, Decision Tree and SVM models.

4. Evaluate the models using regression metrics.
5. Compare the predictive performance of the models.
6. Identify a suitable model for painting cost estimation.

Dataset Description

2.1 Dataset Source

The dataset was obtained from Kaggle [1]. It contains construction and electrical BOQ(Bill of Quantities) information.

Dataset name: Construction and Electrical BOQ Dataset.

Dataset link: https://www.kaggle.com/datasets/kaiwalyaatulraut/construction-and-electrical-boq-dataset

The dataset contains information about construction items, quantities, costs, building types, and project details. For this experiment, the records were filtered to retain painting-related work.

2.2 Dataset Size

After filtering and preprocessing, the dataset used in the experiment contains the following records.

Table 2.1: Dataset and experimental split

Description	Value
Total usable samples	1,199
Unique projects	745
Training samples	846
Testing samples	353
Training projects	521
Testing projects	224
Training features	7

The data set was split into two parts. A project-based split was used to ensure that multiple records relating to the same item would not be split between the test and training sets.

The split means that the training set contains about 70.56% of the usable records whilst the test set contains around 29.44%.

2.3 Input Features and Target Variable

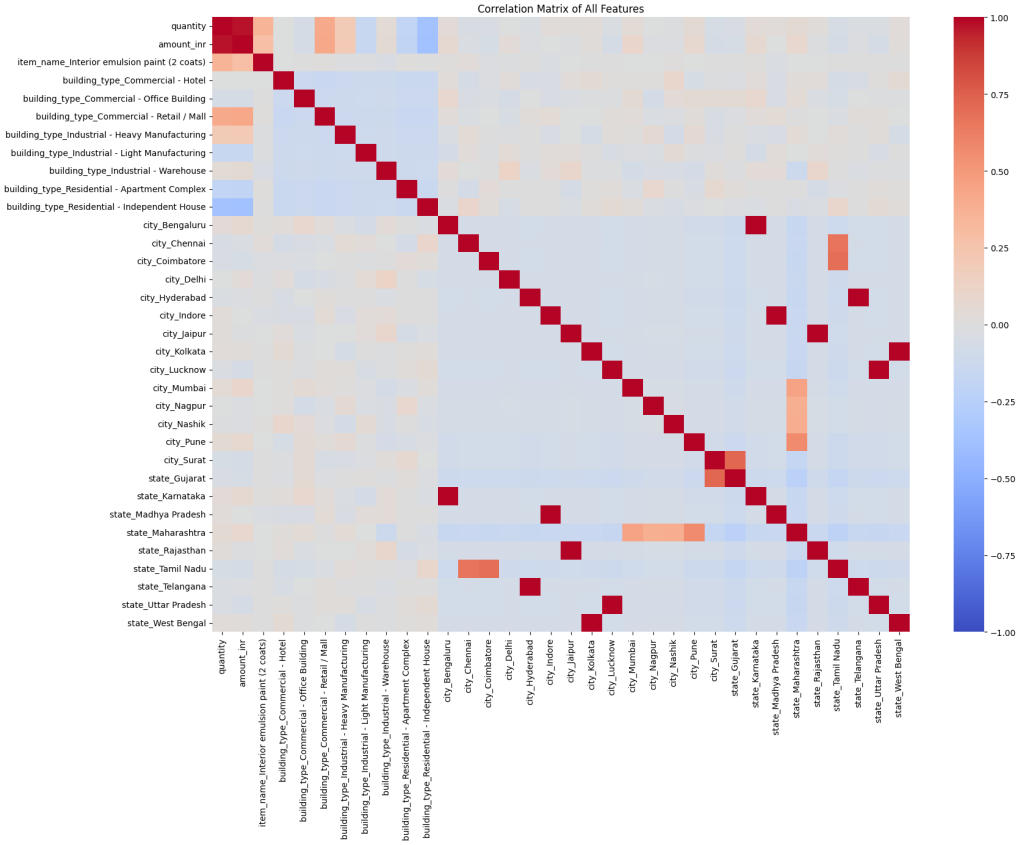


Figure 2.1

The target variable is `amount_inr`, which represents the recorded item amount in Indian rupees.

The input features used in the experiment are listed in Table 2.2.

Table 2.2: Input features and target variable

Feature	**Description**
item_name	Name or type of the painting item.
quantity	Quantity of the painting item in sqm.
building_type	Type of building associated with the project.
city	City where the project is located.
state	State where the project is located.
project_year	Year extracted from the project start date.
project_month	Month extracted from the project start date.
Target: amount_inr	**Recorded amount of the item in INR.**

The original variable `project_start_date` has been converted into year and month variables. This means we will have seven variables to feed into our model:

Categorical variables were converted using one-hot encoding so they can be used by the regression models. Numerical and date derived variables were kept as numerical variables.

2.4 Data Preprocessing

The following preprocessing steps were performed:

1. The relevant dataset files were loaded.
2. The files were merged using the project identifier where required.
3. Records were filtered to retain painting-related items.
4. The project start date was converted into project year and project month.
5. The seven input features and the target variable were selected.
6. Categorical variables were encoded, and the data was divided into training and testing sets by project.

The preprocessing transformations were incorporated into the model pipelines to ensure consistent handling of the input features.

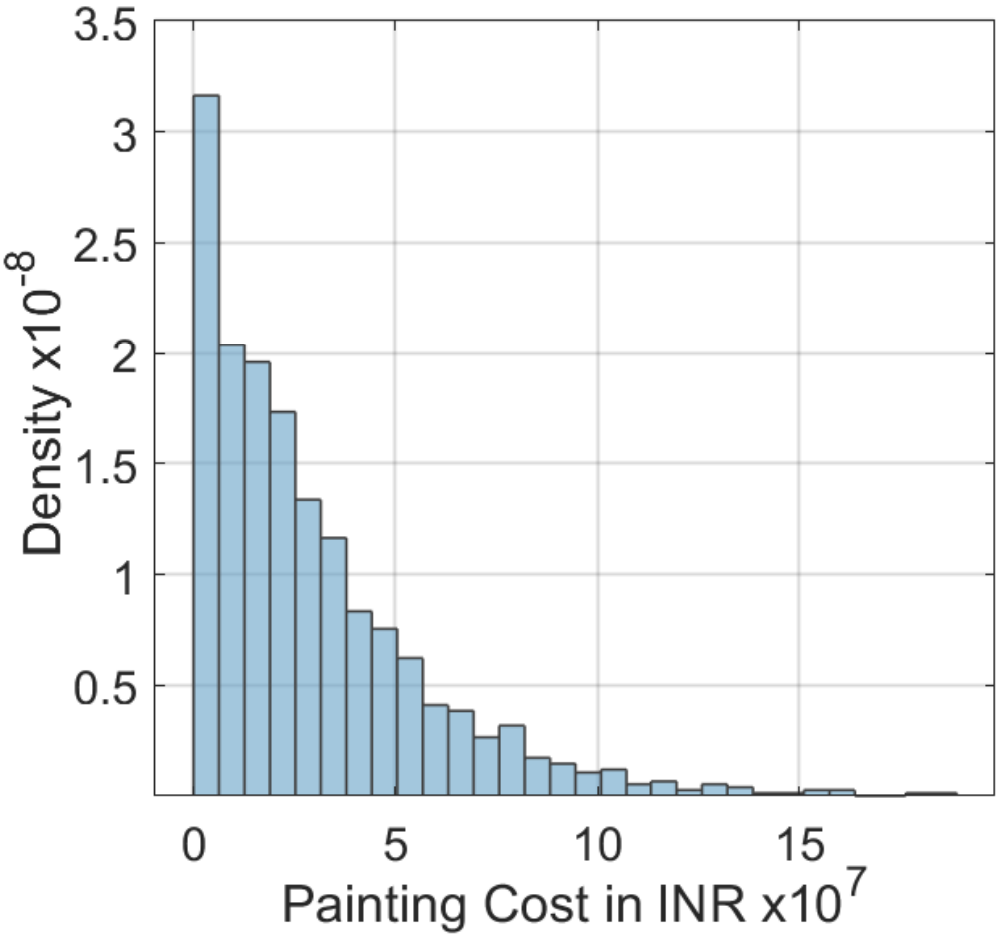


Figure 2.2: Distribution of the painting cost target

Experimental Methodology

3.1 Experimental Setup

Three supervised regression algorithms were used:

- Linear Regression
- Decision Tree Regressor
- Support Vector Machine (Support Vector Regression, SVR)

The training set comprised 846 samples from 521 projects. The test set comprised 353 samples from 224 projects.

Grouped Cross Validation was performed using projects as the groups. This meant that all models were assessed using the same training and testing splits based on projects.

3.2 System Specifications

All experiments were conducted using the following workstation. These specifications were retrieved from the Windows OS.

Table 3.1: Hardware and software specifications of the experimental system

Specification	**Details**
Operating system	Microsoft Windows 11 Pro
OS version/build	Version 10.0.26200, Build 26200
System manufacturer	Gigabyte Technology Co., Ltd.
System model	Z890 EAGLE WIFI7
System architecture	x64-based PC
Processor information	Intel64 Family 6, Model 198, Stepping 2; reported clock speed approximately 3.7 GHz
Installed physical memory (RAM)	97,754 MB

3.3 Linear Regression

Linear Regression attempts to model the relationship between the input variables and the target variable as a linear combination.

In general, this can be modelled as:

$$\hat{y} = \beta_0 + \sum_{j=1}^{p} \beta_j x_j \tag{3.1}$$

Where $\hat{y}$ predicts the cost of a painting, x_j is an input variable, β_j is the coefficient for that variable, β_0 is the intercept, and p is the number of variables.

The coefficients of the model are calculated using ordinary least squares to minimise the sum of square errors between the actual and predicted target variable.

As the price of a painting item is roughly equal to the product of the quantity and rate, this means that the variable will be linear on a log scale. Therefore, in the final pipeline for Linear Regression, we logged the `quantity` variable. We then made the model predict log(target amount) and then transformed our predictions back into rupees using the exponential function. This resulted in a lower MAE on the test data and a better R^2 score than if we had used the raw target, and we can guarantee our predictions will be positive.

3.4 Decision Tree Regressor

A Decision Tree Regressor divides the feature space into different regions using decision rules. Then predictions are made using the value learned for that region.

Decision Trees can capture non-linear relationships and interactions between features.

This model was added to show the difference between the tree-based model and linear model. The model was trained on the raw target variable.

3.5 Support Vector Machine (Support Vector Regression)

Support Vector Regression (SVR) fits a function that deviates from the observed targets by at most a margin ε while remaining as flat as possible. Errors larger than ε are penalized in proportion to their size, controlled by a regularization parameter C.

With a kernel function K, the prediction takes the form

$$\hat{y}(\mathbf{x}) = \sum_{i=1}^{n_{sv}} (\alpha_i - \alpha_i^*)\, K(\mathbf{x}_i, \mathbf{x}) + b \tag{3.2}$$

where $\mathbf{x}_i$ are the support vectors, α_i and α_i are the learned dual coefficients, and b is the bias term. In this project the Radial Basis Function (RBF) kernel was used:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\right) \tag{3.3}$$

The RBF kernel allows the model to capture nonlinear relationships between the features and the cost.

Because SVR is sensitive to feature scale, all numerical features were standardized. As with Linear Regression, the `quantity` feature was log-transformed, and the logarithm of the target was modelled (and standardized) before predictions were converted back to rupees. The hyperparameters C, ε and γ were selected by a grid search using the same project-based grouped cross-validation on the training set, so that no project appears in both the fitting and validation parts of a fold.

3.6 Web Application Implementation

A Streamlit app has been created to interactively predict painting costs. In order to match the final model selection outlined in this report the pipeline will need to be trained using the saved SVR pipeline. The pipeline must also be preprocessed using the same transformations that were applied during training. The app takes inputs from the user and creates a feature vector to be passed into the saved pipeline. The app is deployed and can be found here below. The deployed model will require updating if it still uses the previous Linear Regression pipeline.

The deployed app [2] can be accessed here: https://painting-cost-prediction.streamlit.app/ The full code can be accessed on GitHub [3]: https://github.com/Goutam363/painting-cost-prediction

3.7 Evaluation Metrics

The models were evaluated using MAE, RMSE, and R^2.

3.7.1 Mean Absolute Error

MAE measures the average absolute difference between the actual and predicted values.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{3.4}$$

Here, y_i is the actual value, $\hat{y}_i$ is the predicted value, and n is the number of samples.

A lower MAE indicates a smaller average absolute prediction error.

3.7.2 Root Mean Squared Error

RMSE measures the square root of the average squared prediction error.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3.5}$$

RMSE penalizes large prediction errors more heavily than MAE. A lower RMSE indicates better performance under this metric.

3.7.3 Coefficient of Determination

The coefficient of determination is equal to:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{3.6}$$

where $\bar{y}$ is the mean of the actual target values.

A R^2 value close to 1 generally means that the model explains more of the variation in the target compared to a mean-predictive baseline. It does not exactly reflect the percentage accuracy of prediction.

Results and Analysis

4.1 Cross-Validation Results

Table 4.1 below contains the described cross-validation performance.

Table 4.1: Cross-validation performance

Model	**MAE (INR)**	**RMSE (INR)**	R^2
Linear Regression	3,139,158.91	5,104,909.85	0.9667
Decision Tree	4,951,735.04	8,218,922.69	0.9136
SVM	3299671.90	5551083.50	0.9606

The model with the best cross-validated results is Linear Regression. It has the lowest MAE and RMSE values as well as the highest R^2 value. SVR also performs closely, but Decision Tree has the worst performing cross-validation results. This is an important consideration when choosing which model to go with. Using the test set to decide would mean going with SVR, however using the average cross-validation results would mean going with Linear Regression.

4.2 Final Test Results

The final test results are presented in Table 4.2 below.

Table 4.2: Final test-set performance

Model	**MAE (INR)**	**RMSE (INR)**	R^2
Linear Regression	3,253,582	5,541,855	0.9636
Decision Tree	5,227,014	10,253,605	0.8753
SVM	3,228,139	5,511,006	0.9640

SVR performs the best of all models in terms of the held out test set. This model shows the lowest MAE and RMSE and has a high R^2 score. It performs only marginally better than Linear Regression, showing a reduction in MAE of INR 25,443 (or 0.78%) and a reduction in RMSE of INR 30,849 (or 0.56%). The SVR also shows an increase in R^2 of 0.0004. Both models perform much better than Decision Tree.



Figure 4.1: Comparison of test MAE and RMSE for Linear Regression, Decision Tree and SVM

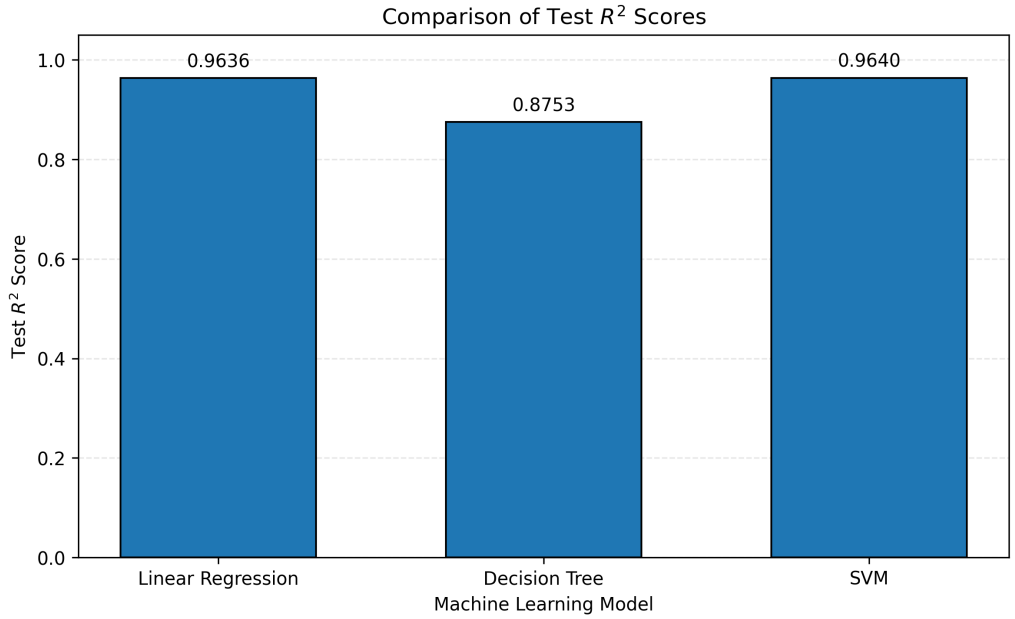


Figure 4.2: Comparison of test R^2 scores for Linear Regression, Decision Tree and SVM

4.3 Actual Versus Predicted Values

We can use the actual vs predicted graph to see how accurate the predictions from each model are. This graph shows actual values on the x-axis and predicted values on the y-axis.

The diagonal line $y = x$ indicates perfect accuracy. The closer the points are to this line the less error there is in the predictions. The further away they are, the more errors there are.

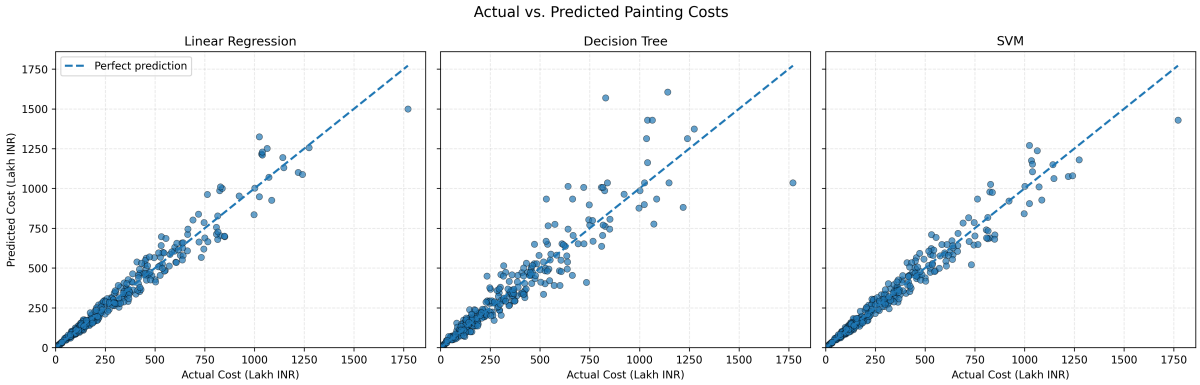


Figure 4.3: Actual versus predicted painting costs on the held-out test set for Linear Regression, Decision Tree and SVM.

The Linear Regression and SVM models predict values that follow the line quite well for most of the range of costs. The Decision Tree predictions are much more spread out, particularly for higher cost items.

4.4 Residual Analysis

A residual is also defined as the difference between an actual value and the predicted value:

$$e_i = y_i - \hat{y}_i \tag{4.1}$$

By analysing the residuals we can find any systematic errors within the predictions. This means that ideally we would like to see the points scattered randomly around the horizontal axis. If the residuals are large then this suggests that it was very difficult to predict this point. We may want to remove this point from our data set or improve our model.

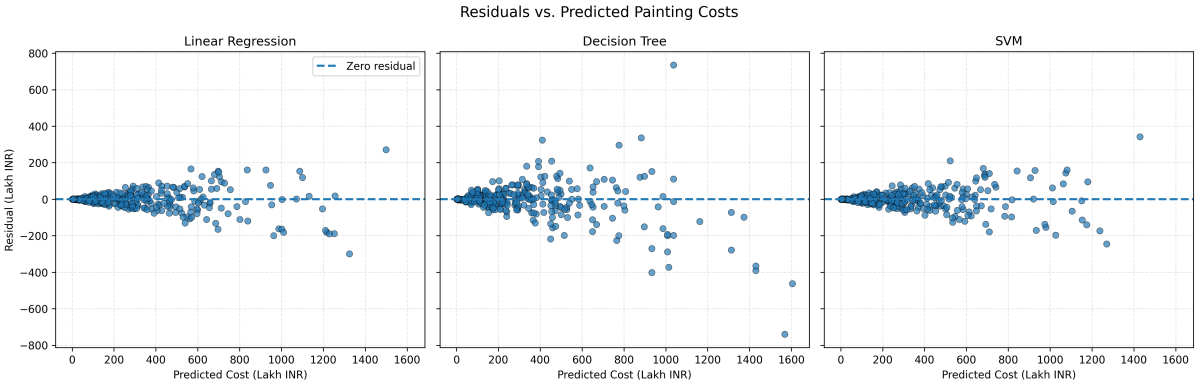


Figure 4.4: Residuals versus predicted painting costs for Linear Regression, Decision Tree and SVM

4.5 Discussion

Overall both SVR and Linear Regression are significantly better than Decision Tree. There is an overall slight bias towards Linear Regression in the cross validation table. However, looking at the final test set, SVR has performed better for all 3 metrics. Due to the small difference between the two models I don't think we can really say one outperforms the other, however as we're using the metrics from the test set to choose our best performing model, SVR is chosen.

Conclusion

3 models were tested for the painting cost estimation model using Linear Regression, Decision Tree and Support Vector Regression (SVR)

We can see that SVR had the best results out of the 3 models when tested on the validation set with an MAE of INR 3,228,139, RMSE of INR 5,511,006 and an R^2 of 0.9640. Linear Regression had a slightly higher MAE and RMSE but still decent scores of INR 3,253,582 and INR 5,541,855 with an R^2 of 0.9636. Decision Tree had the worst results of the 3 models with an MAE of INR 5,227,014, RMSE of INR 10,253,605 and an R^2 of 0.8753.

Overall with the final testing results we can conclude that the SVR model will be used for our painting cost estimation. This is because it has the lowest MAE and RMSE scores and highest R^2.

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[2] Painting Cost Estimation Project Team, "Building painting cost estimation: Streamlit web application," https://painting-cost-prediction.streamlit.app/, 2026, online application.

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